

# Computer Network and Communication CNS3200

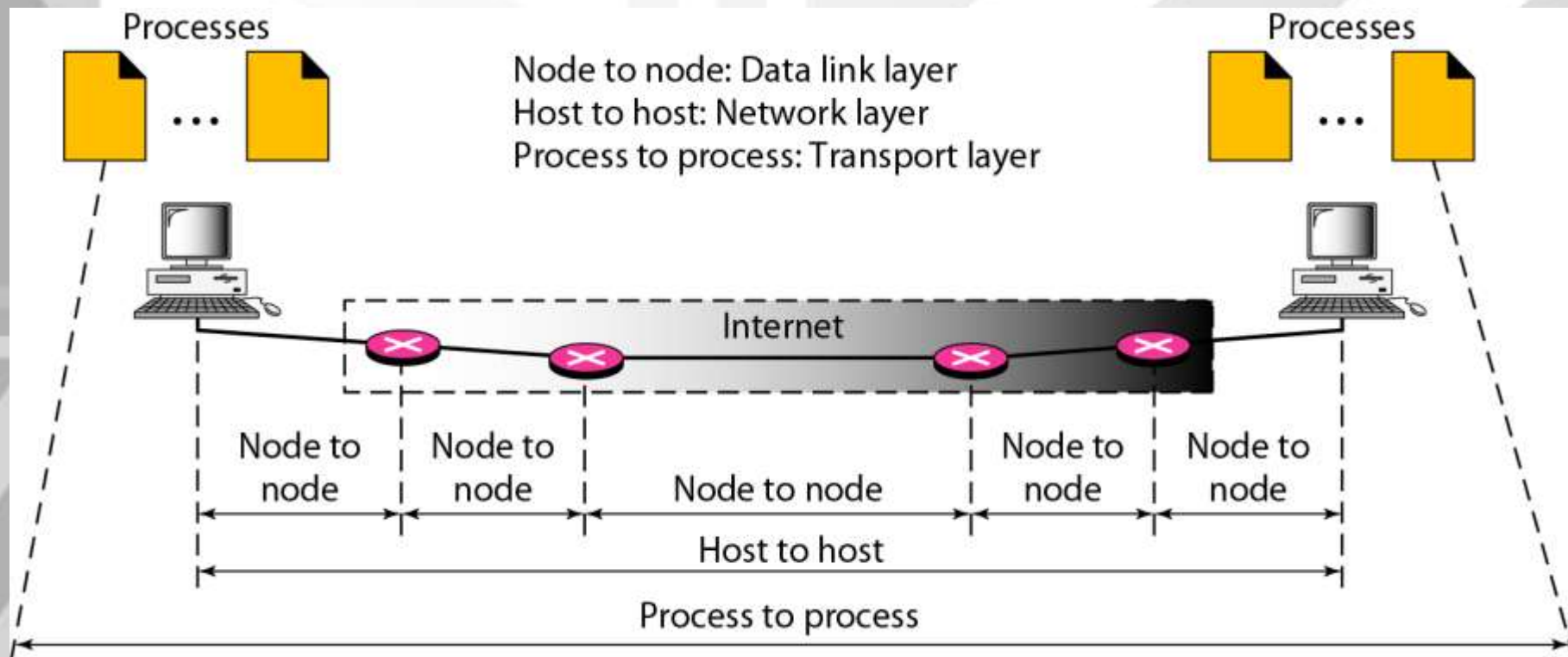
# Learning Outcome

- **EXPLAIN FUNCTION OF TRANSPORT LAYER AND RELATED PROTOCOLS (C4)**
- **DIFFERENTIATE BETWEEN UDP AND TCP PROTOCOL (C4)**
- **EXPLAIN SUITABLE METHOD FOR TRAFFIC CONTROL (C3)**

# TCP and UDP Protocols

- ❑ *TCP and UDP are the transport layer protocols*
- ❑ *The transport layer is responsible **for process-to-process delivery**—the delivery of a segment, part of a message, from one process to another. Two processes communicate in a client/server relationship, as we will see later.*
- ❑ *These protocols use **port numbers** to accomplish this.*
- ❑ *Port numbers provide end-to-end addresses at the transport layer and allow multiplexing and demultiplexing at this layer, just as IP addresses do at the network layer. Table 24.1 gives some common port numbers for all three protocols we discuss in this chapter.*

Figure 23.1 *Types of data deliveries*



**Table 24.1:** Some well-known ports used with UDP and TCP



| <i>Port</i> | <i>Protocol</i> | <i>UDP</i> | <i>TCP</i> | <i>Description</i>                     |
|-------------|-----------------|------------|------------|----------------------------------------|
| 7           | Echo            | √          |            | Echoes back a received datagram        |
| 9           | Discard         | √          |            | Discards any datagram that is received |
| 11          | Users           | √          | √          | Active users                           |
| 13          | Daytime         | √          | √          | Returns the date and the time          |
| 17          | Quote           | √          | √          | Returns a quote of the day             |
| 19          | Chargen         | √          | √          | Returns a string of characters         |
| 20, 21      | FTP             |            | √          | File Transfer Protocol                 |
| 23          | TELNET          |            | √          | Terminal Network                       |
| 25          | SMTP            |            | √          | Simple Mail Transfer Protocol          |
| 53          | DNS             | √          | √          | Domain Name Service                    |
| 67          | DHCP            | √          | √          | Dynamic Host Configuration Protocol    |
| 69          | TFTP            | √          |            | Trivial File Transfer Protocol         |
| 80          | HTTP            |            | √          | Hypertext Transfer Protocol            |
| 111         | RPC             | √          | √          | Remote Procedure Call                  |
| 123         | NTP             | √          | √          | Network Time Protocol                  |
| 161, 162    | SNMP            |            | √          | Simple Network Management Protocol     |

Figure 23.2 *Port numbers*

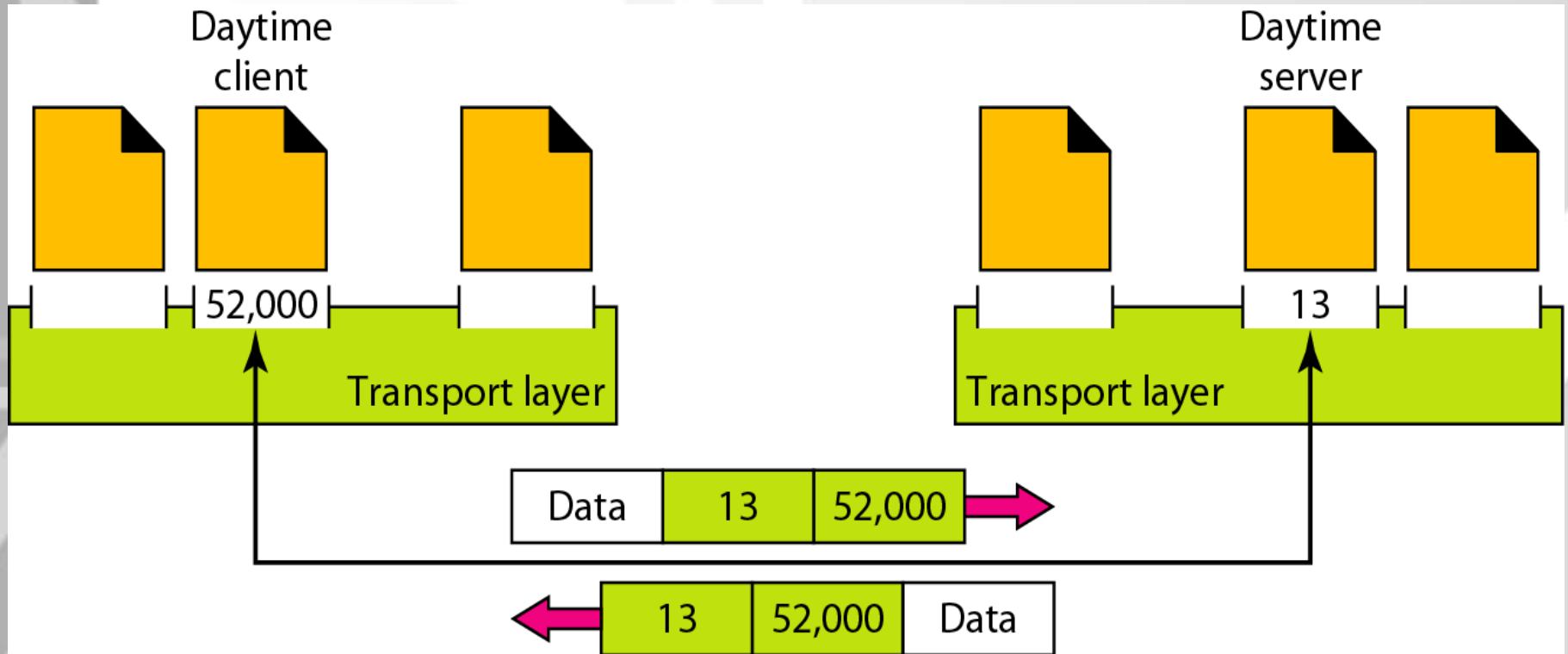


Figure 23.3 *IP addresses versus port numbers*

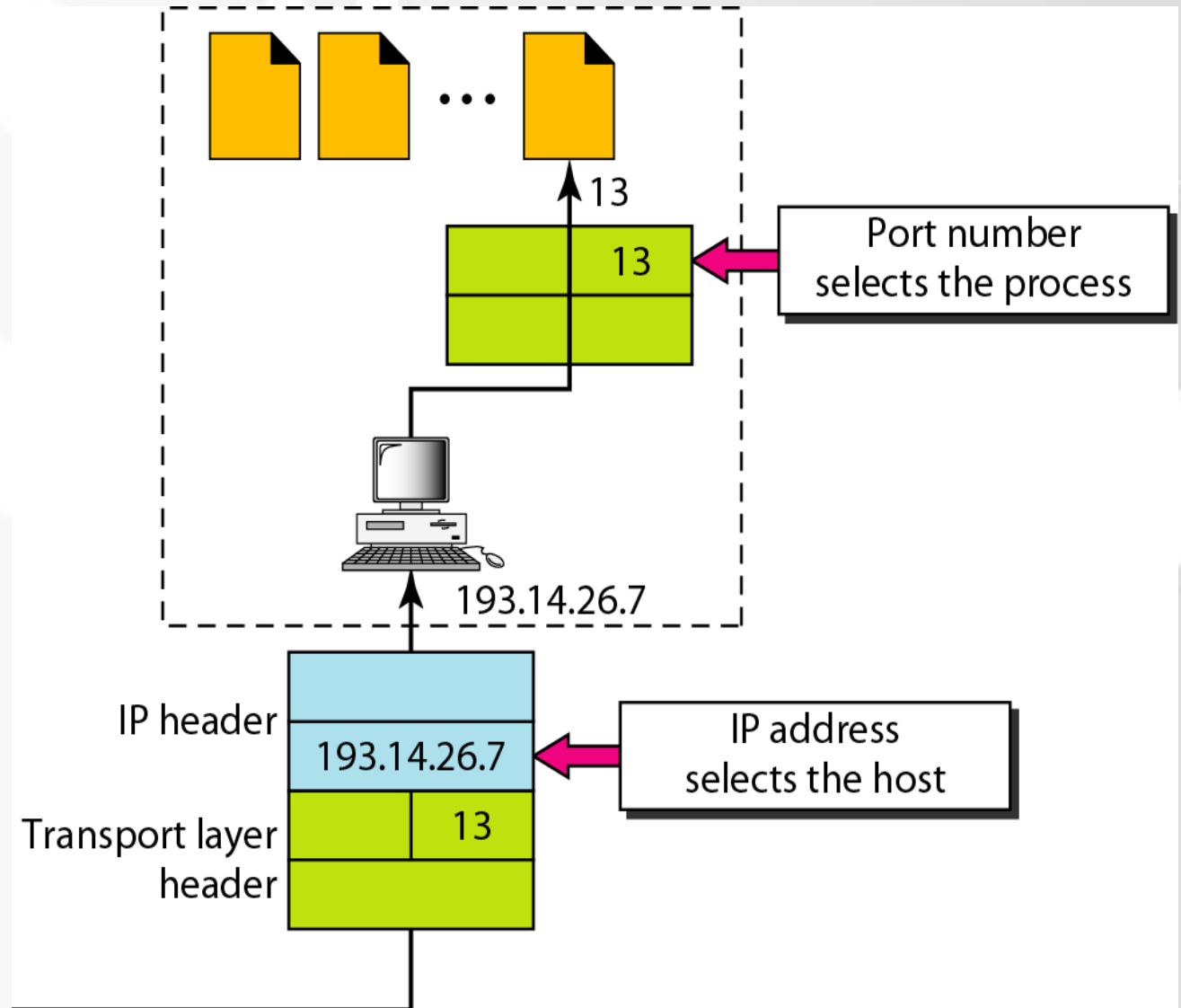
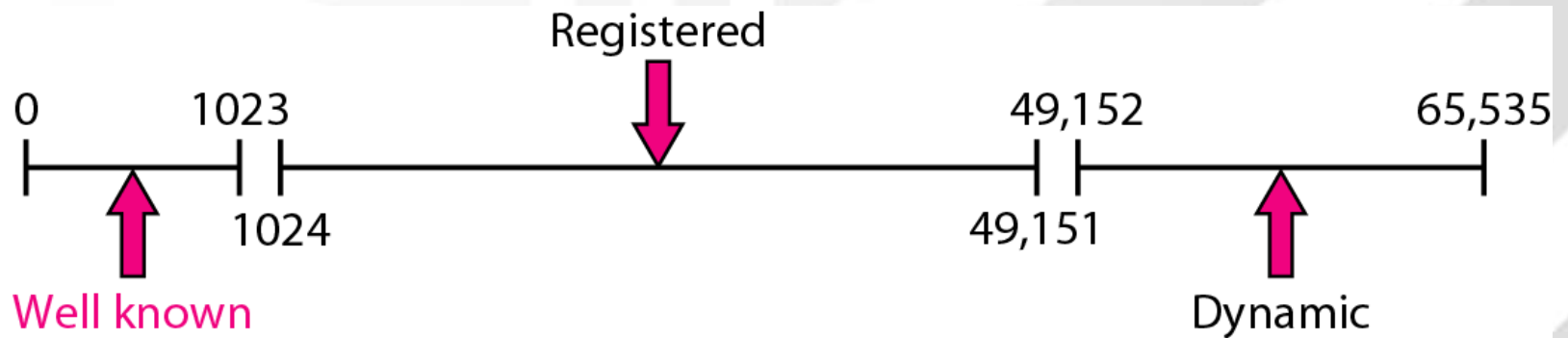




Figure 23.4 *IANA ranges*



IANA: Internet Assigned Number Authority



Figure 23.5 *Socket address*

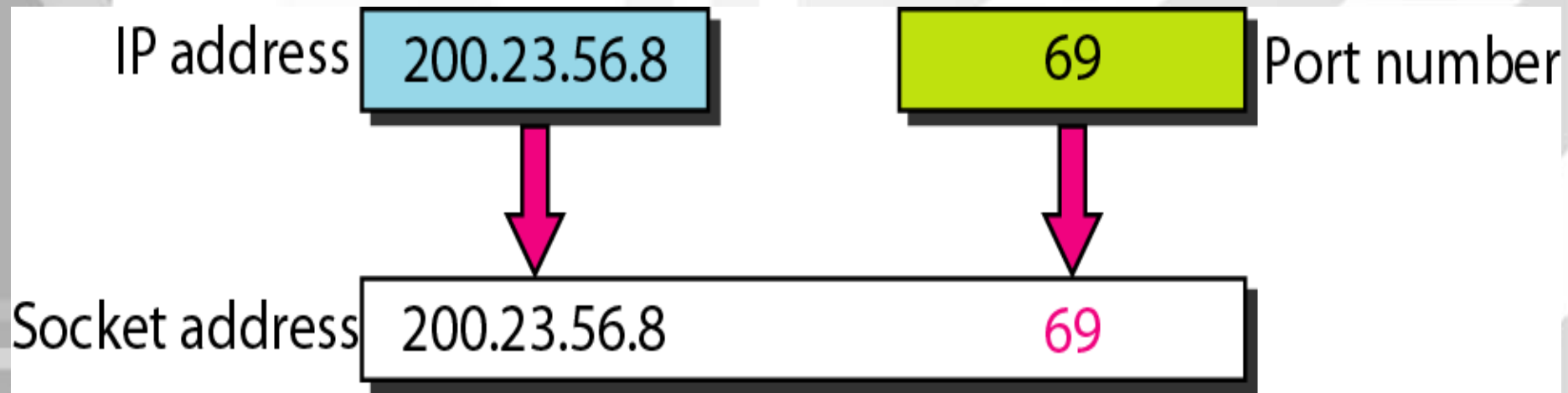


Figure 23.6 *Multiplexing and demultiplexing*

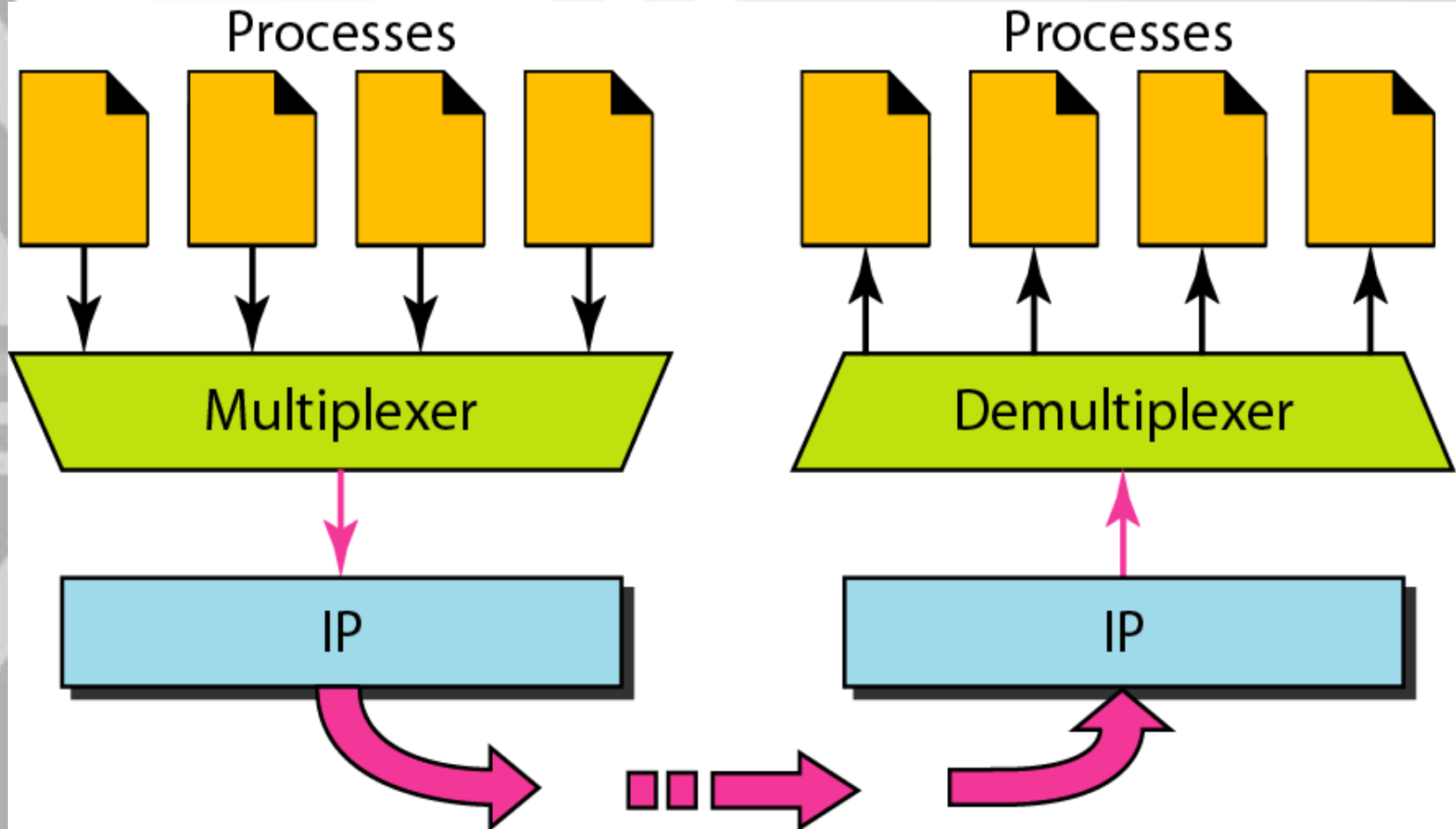


Figure 23.7 *Error control*

- Error is checked in these paths by the data link layer
- Error is not checked in these paths by the data link layer

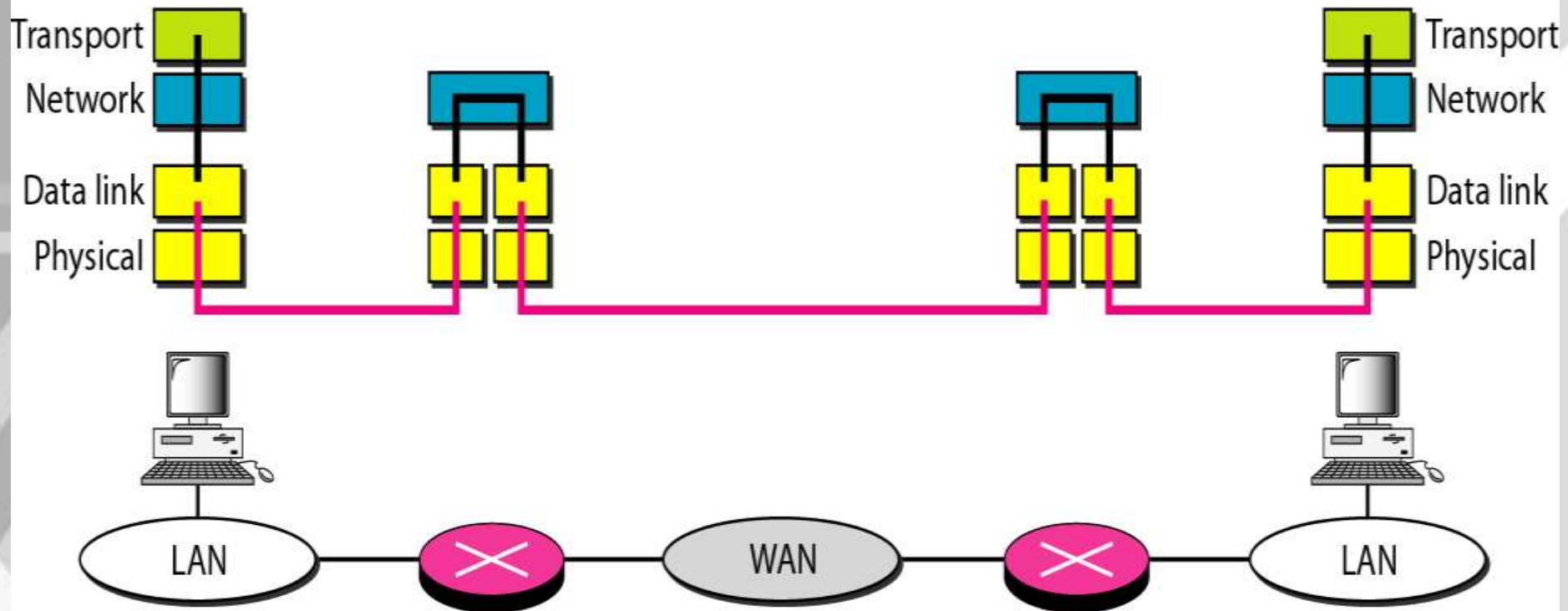
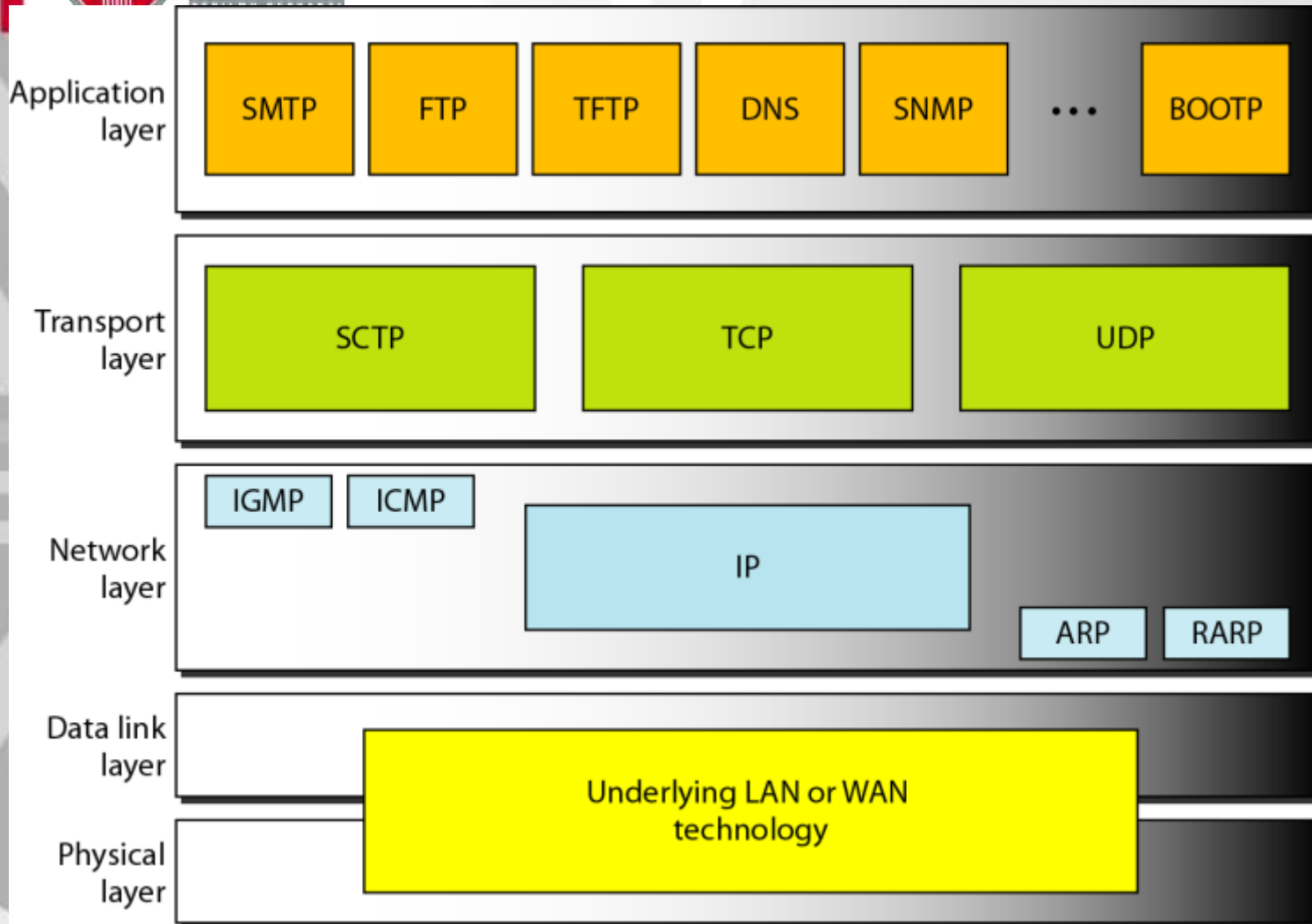


Figure 23.8 *Position of UDP and TCP in TCP/IP suite*



# TCP vs UDP

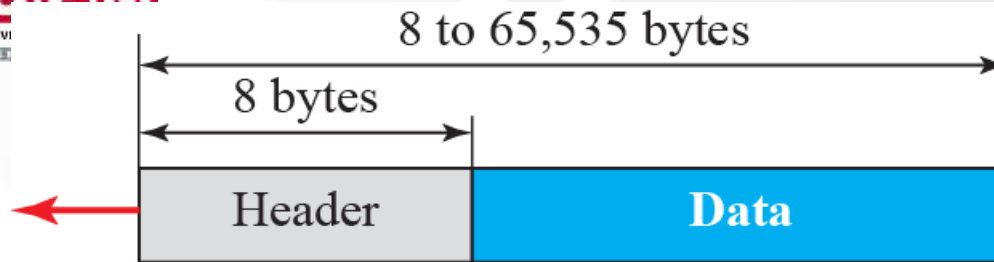
| Transport Layer Functions      | TCP (reliable)      | UDP (unreliable) |
|--------------------------------|---------------------|------------------|
| Process to process delivery    | √                   | √                |
| Connection control             | Connection-Oriented | Connectionless   |
| Flow control (Sliding window)  | √                   | -                |
| Error Control (Retransmission) | √                   | -                |

## 24.2.1 User Datagram

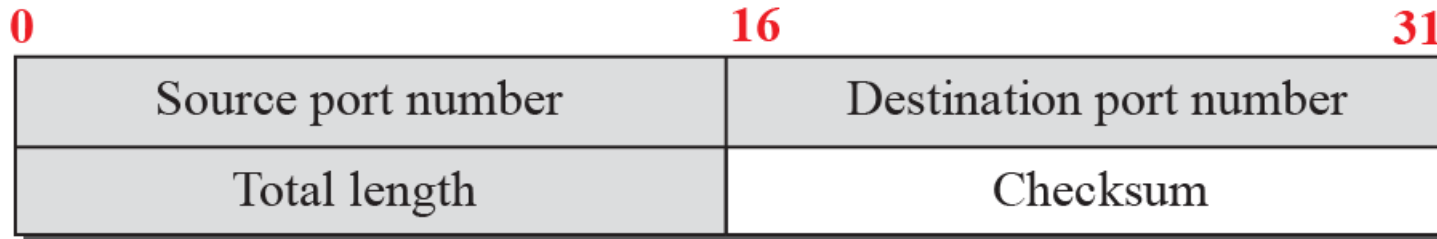


*UDP packets, called user datagrams, have a fixed-size header of 8 bytes made of four fields, each of 2 bytes (16 bits). Figure 24.2 shows the format of a user datagram. The first two fields define the source and destination port numbers. The third field defines the total length of the user datagram, header plus data. The 16 bits can define a total length of 0 to 65,535 bytes. However, the total length needs to be less because a UDP user datagram is stored in an IP datagram with the total length of 65,535 bytes. The last field can carry the optional checksum (explained later).*

**Figure 24.2: User datagram packet format**



a. UDP user datagram



b. Header format





The following is the contents of a UDP header in hexadecimal format.

**CB84000D001C001C**

- a.** What is the source port number?
- b.** What is the destination port number?
- c.** What is the total length of the user datagram?
- d.** What is the length of the data?
- e.** Is the packet directed from a client to a server or vice versa?
- f.** What is the client process?

## Solution



- a. The source port number is the first four hexadecimal digits  $(CB84)_{16}$  or 52100
- b. The destination port number is the second four hexadecimal digits  $(000D)_{16}$  or 13.
- c. The third four hexadecimal digits  $(001C)_{16}$  define the length of the whole UDP packet as 28 bytes.
- d. The length of the data is the length of the whole packet minus the length of the header, or  $28 - 8 = 20$  bytes.
- e. Since the destination port number is 13 (well-known port), the packet is from the client to the server.
- f. The client process is the Daytime (see Table 3.1).

## 24-3 TCP

***Transmission Control Protocol (TCP) is a connection-oriented, reliable protocol. TCP explicitly defines connection establishment, data transfer, and connection teardown phases to provide a connection-oriented service.***

## 24.3.8 Error Control



*TCP is a reliable transport-layer protocol. This means that an application program that delivers a stream of data to TCP relies on TCP to deliver the entire stream to the application program on the other end in order, without error, and without any part lost or duplicated.*

## *24.3.9 TCP Congestion Control*

*TCP uses different policies to handle the congestion in the network. We describe these policies in this section.*



## **UDP: Anything where you don't care too much if you get all data always**

- Tunneling/VPN (lost packets are ok - the tunneled protocol takes care of it)
- Media streaming (lost frames are ok)
- Games that don't care if you get every update
- Local broadcast mechanisms (same application running on different machines "discovering" each other)

## **TCP: Almost anything where you have to get all transmitted data**

- Web
- SSH, FTP, telnet
- SMTP, sending mail
- IMAP/POP, receiving mail

## ***Examples of application use the UDP service:***

1. ***DNS***
2. ***Trivial File Transfer Protocol***
3. ***Multicasting***
4. ***Simple Network Management Protocol***
5. ***Routing Information Protocol***

***Stream Control Transmission Protocol (SCTP) is a new transport-layer protocol designed to combine some features of UDP and TCP in an effort to create a protocol for multimedia communication.***